3.7 ADULTS

Gastrointestinal Infections

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Infectious Diarrhoea			
Acute mild to moderate diarrhoea Virus: Norovirus (usually) Bacteria (Salmonella) Parasite (rarely)	To replace fluids. No antibiotics required.		• *It is usually self-limiting, requiring supportive therapy. • Antibiotics are indicated in severe disease or immunocompromised patients. • Send stool culture & blood culture prior to starting antibiotics. • Targeted therapy is based on culture and sensitivity. • Convert to oral once clinically improved. • Duration will be based on diagnosis or organism isolated. • **Mild:** ≤ 3 unformed stools, minimal symptoms • **Moderate:** ≥ 3 unformed stools and/or symptomatic • **Severe:** ≥ 6 unformed stools/day +/- fever (>38.5°C), faecal

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			leukocytes, bloody stools or tenesmus.
Acute severe diarrhoea Campylobacter Yersinia *Salmonella enterica* Aeromonas *Plesiomonas sp* *Shigella shigelloides*	**Empirical treatment***: Ciprofloxacin 500 mg - 750 mg PO q12H for 3 - 5 days	Ceftriaxone 2 g IV q24H **OR** Azithromycin 500 mg PO q24H for 3 days **OR** Azithromycin 1 g PO single dose **OR** Co-trimoxazole (480 mg) 2 tablets PO q12H	• *In immunocompetent adults, empiric antimicrobial therapy is usually not indicated, except for the following: • Ill, immunocompetent patient with fever documented in a medical setting, abdominal pain, bloody diarrhoea, and bacillary dysentery (frequent scanty bloody stools, fever, abdominal cramps, tenesmus) presumptively due to Shigella. • People who have recently travelled internationally with body temperatures ≥38.5°C and/or signs of sepsis • Suspected enteric fever
Vibrio cholerae	Antibiotics therapy is not required in mild disease. Primary treatment is hydration. **In severe disease or immunocompromised patient:**		
	Azithromycin 1 g PO single dose	Doxycycline 300 mg PO single dose	
Clostridium difficile			• *As an alternative when oral Vancomycin is not available.

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Initial, non-severe	Vancomycin 125 mg PO q6H for 10 days	Metronidazole 400 mg PO q8H for 10 days*	• Discontinue therapy with inciting antibiotic agent as soon as possible as this may influence *Clostridium difficile* infection recurrence. • If persistent/recurrent, to refer to the ID team.
Initial, severe (WBC > $15x10^9$/L, creatinine 50% increase from baseline, temperature >38.5°C, clinical evidence of severe colitis)	Vancomycin 125 mg PO q6H + Metronidazole 500 mg IV q8H		• Duration: 10 days.
Fulminant disease (Hypotension or shock, ileus, megacolon)	**Absence of ileus:** Vancomycin 500 mg PO or via nasogastric tube 4 times daily + Metronidazole 500 mg IV q8H **If ileus is present, additional**		• If fulminant disease, refer to the ID team and consult a surgeon. • Duration: 10 days.

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	considerations include: Faecal microbiota transplant (administered rectally) **OR** Rectal Vancomycin (administered as a retention enema 500 mg in 100 mL normal saline per rectum; retained for as long as possible and readministered every 6 hours)		

Prophylaxis Against Travellers' Diarrhoea (TD)

1. Antibiotics are **NOT** recommended.
2. Prophylactic antibiotics afford no protection against non-bacterial pathogens and can remove normally protective micro-flora from the bowel, increasing the risk of infection with resistant bacteria pathogens.
3. Travellers may become colonised with ESBL–producing bacteria.
4. Prophylactic antibiotics limit the therapeutic options if TD occurs.

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
Spontaneous Bacterial Peritonitis (SBP)			
Community-acquired *E. coli* *Klebsiella pneumoniae* *Staphylococcus aureus* *Enterococcus faecalis* *Enterococcus faecium*	**No history of antibiotic exposure:** Co-amoxiclav 1.2 g IV q8H **History of antibiotic exposure:** Ceftriaxone 2 g IV q24H		• Empiric antibiotic therapy for SBP is indicated for patients with ascites who have any of the following: • Temperature greater than 37.8°C (100°F) • Abdominal pain and/or tenderness • Altered mental status • Ascitic fluid polymorphonuclear leukocyte (PMN) count ≥250 cells/microL • Peritoneal fluid is preferably sent prior to starting empirical antibiotics to guide antibiotics choice. • Duration: 5 to 7 days.
Hospital-acquired Occurs >48 hours after hospital admission	**Hospital-acquired infection without instrumentation:** Piperacillin/Tazobactam 4.5 g IV q6H		
Secondary bacterial peritonitis Post-instrumentation	Piperacillin/Tazobactam 4.5 g IV q6H ±		

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Vancomycin 15 mg/kg (max 1 g) IV q8-12H		
Perforated Gastric/Duodenal Ulcer			
Community-acquired	Co-amoxiclav 1.2 g IV q8H	Cefuroxime 750-1500 mg IV q8H + Metronidazole 500 mg IV q8H	
Hospital-acquired	**If patient is in shock:** Piperacillin/Tazobactam 4.5 g IV q6H **If patient is not in shock:** Ceftazidime 2 g IV q8H **OR** Cefepime 2 g IV q8H +		

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Metronidazole 500 mg IV q8H		
Hepatic Abscess			
Pyogenic abscess Enteric gram-negative bacilli, particularly *E. coli* and *K. pneumoniae* *Streptococcus milleri* group (including *Streptococcus anginosus*, *Streptococcus constellatus*, and *Streptococcus intermedius*) **Need to search for simultaneous infection at other loci promptly.**			
Community-acquired	Co-amoxiclav 1.2 g IV q8H	Ceftriaxone 2 g IV q24H + Metronidazole 500 mg IV q8H	• Abscess drainage is the optimal therapy for pyogenic liver abscesses. • Ensure blood cultures are taken and de-escalate antibiotics accordingly. • Total antibiotic duration: 14 - 42 days depending on response to therapy. • Longer courses (up to several months) may be required in the patient who is inadequately drained or treated without drainage. • IV antibiotics should be switched to oral once clinical improvement occurs (afebrile >48 hours, adequate clinical response). • If stepping down to oral Co-amoxiclav, give a higher dose of 1
Hospital-acquired (secondary infection) Post-invasive procedures involving GIT such as ERCP, etc.	Piperacillin/Tazobactam 4.5 g IV q6H		

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
			g q8H.
Amoebic abscess *Entamoeba histolytica*	**Tissue agent:** Metronidazole 800 mg PO q8H for 7-10 days **Luminal agent to eliminate residual colonic colonization:** Paromomycin* 500 mg PO q8H for 7 days	**Tissue agent:** Tinidazole* 800 mg PO q8H **OR** Tinidazole* 2 g PO q24H for 3 - 5 days **Luminal agents:** Iodoquinol* 650 mg PO q8H for 20 days **OR** Diloxanide Furoate* 500 mg PO q8H for 10 days	• *Tinidazole, Paromomycin, Iodoquinol, and Diloxanide furoate are not available in Malaysia.
***Helicobacter pylori* Infection**			
First-Line Therapy	Proton pump inhibitor* PO q12H +	**Penicillin Allergy:** Proton pump inhibitor* PO q12H	• *Proton pump inhibitors: • Omeprazole 20 mg PO q12H

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Amoxicillin 1 g PO q12H + Clarithromycin 500 mg PO q12H **OR** Metronidazole*** 400 mg PO q12H	+ Bismuth preparation** + Tetracycline** 500 mg PO q6H + Clarithromycin 500 mg q12H **OR** Metronidazole*** 400 mg PO q8H	• Pantoprazole 40 mg PO q12H • Esomeprazole 20 mg PO q12H • **Not available in UMMC (available in UMSC). • ***Consider Metronidazole if Clarithromycin resistance rate >15%. • Duration: 14 days.
Second-Line Therapy* **	**Bismuth quadruple regime** Proton pump inhibitor* PO q12H + Bismuth preparation** + Tetracycline** 500 mg PO q6H +		• *Proton pump inhibitors: • Omeprazole 20 mg PO q12H • Pantoprazole 40 mg PO q12H • Esomeprazole 20 mg PO q12H • **Not available in UMMC (available in UMSC). • ***Consider second-line therapy if Clarithromycin resistance rate >15%.

CHAPTER 3.7	GASTROINTESTINAL INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
	Metronidazole 400 mg PO q8H		• Duration: 14 days.
	Fluoroquinolone triple therapy Proton pump inhibitor* PO q12H + Levofloxacin 500 mg PO q24H + Amoxicillin 1 g PO q12H		
	Rescue therapy (High dose dual therapy) Amoxicillin 1 g PO q6-8H + Proton pump inhibitor* PO q8H		

References:

1. Randel, Amber. "H. pylori infection: ACG updates treatment recommendations." American family physician 97.2 (2018): 135-137.
2. Goh KL, Manikam J, Qua CS. High-dose rabeprazole-amoxicillin dual therapy and rabeprazole triple therapy with amoxicillin and levofloxacin for 2 weeks as first and second line rescue therapies for Helicobacter pylori treatment failures. Aliment Pharmacol Ther. 2012 May;35(9):1097-102. doi: 10.1111/j.1365-2036.2012.05054.x. Epub 2012 Mar 8. PMID: 22404486.
3. Goh, K- L, Chang, J, Leow, AH- R. High- dose proton pump inhibitor and amoxicillin dual therapy with or without bismuth for 14 days as rescue therapies after Helicobacter pylori treatment failure. J Dig Dis. 2020; 21: 566– 570. https://doi.org/10.1111/1751-2980.12929
4. Hwong-Ruey Leow A, Chang JV, Goh KL. Searching for an optimal therapy for H pylori eradication: High-dose proton-pump inhibitor dual therapy with amoxicillin vs. standard triple therapy for 14 days. Helicobacter. 2020 Oct;25(5):e12723. doi: 10.1111/hel.12723. Epub 2020 Jul 26. PMID: 32713104.
5. Kelly, C. P., Lamont, T., & Bakken, J. S. (2024, March 19). Clostridioides difficile infection in adults: Treatment and prevention. UpToDate.